

DNA DAMAGE AND REPAIR

DNA damage may lead to mutations (changes in DNA sequence). The ability of cells to repair DNA damage and maintain genome stability is of utmost importance to prevent malignant transformation. Different agents induce different types of DNA damage, which in turn require specific responses and repair pathways. A number of hereditary disorders are characterized by genome instability due to defects in genes involved in DNA repair or DNA damage signaling. Many laboratory tests can be used to diagnose genome instability and/or DNA repair defects. Inherited or acquired genome instability is associated with an increased cancer risk.

UV-Induced Mutations

Substitution mutations are typical of UV light exposure and are rarely induced by other mutagens. These have therefore been termed **UV-signature mutations**.

Types of DNA Damage by Agent

TABLE 109-1 Cellular Damage Induced by Physical and Chemical Agents

AGENT	DAMAGE
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Ultraviolet radiation	Dipyrimidine cyclobutane dimers (TT, TC, CT, or CC), pyrimidine-pyrimidone (6-4) photoproducts (mostly TC), DNA-protein cross-links
X-irradiation	DNA single- and double-strand breaks, oxidative base damage
Psoralens plus ultraviolet A	DNA-psoralen monoadducts, DNA interstrand cross-links (binds to T at TA sequences)
Mitomycin C	DNA interstrand cross-links
Benzo-[a]-pyrene	Bulky adducts
Reactive oxygen species	Oxidative base damage (8-oxo-deoxyguanine, thymine glycol), cyclopurines (A or G) forming bulky lesions

DNA Repair Pathways by Damage Type

TABLE 109-2 Types of DNA Damage and Associated DNA Repair Pathways

TYPE OF DNA DAMAGE	DNA REPAIR PATHWAY
DNA photoproducts (CPDs, 6,4-PP)	Nucleotide excision repair
Oxidative base modifications (e.g., 8-oxoG)	Base excision repair
Incorrect DNA base pairing	Mismatch repair
DNA double-strand breaks	Non-homologous end joining, homologous recombination
DNA adducts	Nucleotide excision repair
DNA cross-links (interstrand and intrastrand)	Recombination repair
Persistent DNA lesions	Translesion (bypass) DNA synthesis

CPDs = cyclobutane pyrimidine dimers; 6,4-PP = 6,4-pyrimidine-pyrimidone photoproducts; 8-oxoG = 8-oxo-deoxyguanine.

Cellular Defense Mechanisms Against DNA Damage

DNA repair is a critical cellular defense mechanism that prevents mutation formation at sites of DNA damage after UV exposure. However, it is not the sole protective strategy.

- **Cell cycle arrest:** DNA damage can trigger a temporary halt in cell cycling, providing additional time for repair before replication.
- **Apoptosis:** Programmed cell death eliminates cells with severe DNA damage, reducing the likelihood of propagating mutations.
- **Melanogenesis and epidermal thickening:** Increased melanin production and thickening of the epidermis and stratum corneum reduce penetration of future UV radiation.
- **Immune surveillance:** The host immune system helps remove mutated cells, limiting carcinogenic potential.

These mechanisms collectively minimize the risk of malignant transformation following DNA damage.

More than 100 DNA repair genes have been identified (http://www.cgal.icnet.uk/DNA_Repair_Genes.html#NER).